

MediTrust

Sprint 2 Retrospective

Group7

1. What Went Well

Sprint 2 went well overall, and we were able to complete all planned tasks within the sprint timeline.

During this sprint, we focused on integrating a machine learning model into the MediTrust system. Ravi mainly worked on the machine learning pipeline, backend API integration, and Git branch management. Uday worked on improving the frontend risk assessment interface and connecting it with the prediction API.

This clear division of responsibilities helped both of us work in parallel without major conflicts.

We successfully completed the following:

- Implemented data preprocessing and model training pipeline
- Integrated the trained ML model with the FastAPI backend
- Developed a prediction API endpoint for risk assessment
- Improved the frontend risk assessment page with clinical parameters
- Connected frontend input with backend prediction API
- Verified the full workflow from frontend to ML prediction

By the end of Sprint 2, the system was able to generate cardiovascular risk predictions using a trained machine learning model instead of the simple rule-based logic used in Sprint 1.

2. What Didn't Go Well

Although the sprint was completed successfully, we faced a few technical challenges during development.

One issue occurred while merging Git branches. At one point, the frontend updates were not appearing in the sprint branch because the local repository was not updated with the latest remote commits. We resolved this by fetching the latest changes and merging the branch again.

Another challenge involved aligning the frontend input parameters with the features expected by the machine learning model. Some field names and numeric formats had to be adjusted before the backend could process the input correctly.

These issues required additional debugging and testing but were eventually resolved.

3. What We Learned

From this sprint, we learned several important things:

- How to train machine learning models using healthcare datasets
- How to preprocess data before model training
- How to integrate ML models with a FastAPI backend
- How frontend and backend communicate using REST APIs
- How Git feature branches help manage collaborative development

This sprint improved our understanding of building AI-powered web applications that combine machine learning with full-stack development.

4. What Can Be Improved

For the next sprint, we plan to improve the system in several ways.

We would like to enhance the system by adding explainability features so users can understand why the model predicts a certain risk level. We also want to improve the frontend interface to make the prediction results easier to interpret. Also like to split single front end file to multiple files.

Other improvements include:

- Adding better input validation on the frontend
- Improving model evaluation and accuracy
- Adding visualizations for prediction results
- Expanding testing of the full system workflow

These improvements will help make the system more reliable and user-friendly.

5. Team Member Contributions

Ravi

- Implemented machine learning training pipeline
- Developed preprocessing and model evaluation scripts
- Integrated ML model with FastAPI backend
- Created prediction API endpoint
- Managed Git branching and sprint merges
- Prepared Sprint 2 documentation

Uday

- Improved frontend risk assessment interface
- Implemented patient clinical parameter input form
- Connected frontend form with backend prediction API
- Assisted in testing the full prediction workflow

Both team members collaborated during system testing and final verification.

6. Overall Reflection

Sprint 2 significantly advanced the MediTrust project.

The system now integrates:

Frontend → FastAPI Backend → Machine Learning Model → Risk Prediction

This sprint transformed MediTrust from a simple prototype into a more realistic AI-enabled healthcare prediction system.

The project is now ready for further improvements in Sprint 3, where we plan to add explainability features and improve the overall user experience.